

SUPPLEMENTAL MATERIALS

ASCE Journal of Water Resources Planning and Management

Battle of Water Demand Forecasting

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The Battle of Water Demand Forecasting (BWDF)

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- **Figure S6.** Boxplot of the single rank values if evaluation weeks are considered individually: (a) W1, (b) W2, (c) W3 and (d) W4.
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- **Figure S8.** Boxplot of the single rank values if DMAs are considered individually (part II): (a) DMA 5, (b) DMA 6, (c) DMA 7 and (d) DMA 8.
- **Figure S9.** Boxplot of the single rank values if DMAs are considered individually (part III): (a) DMA 9 and (b) DMA 10.

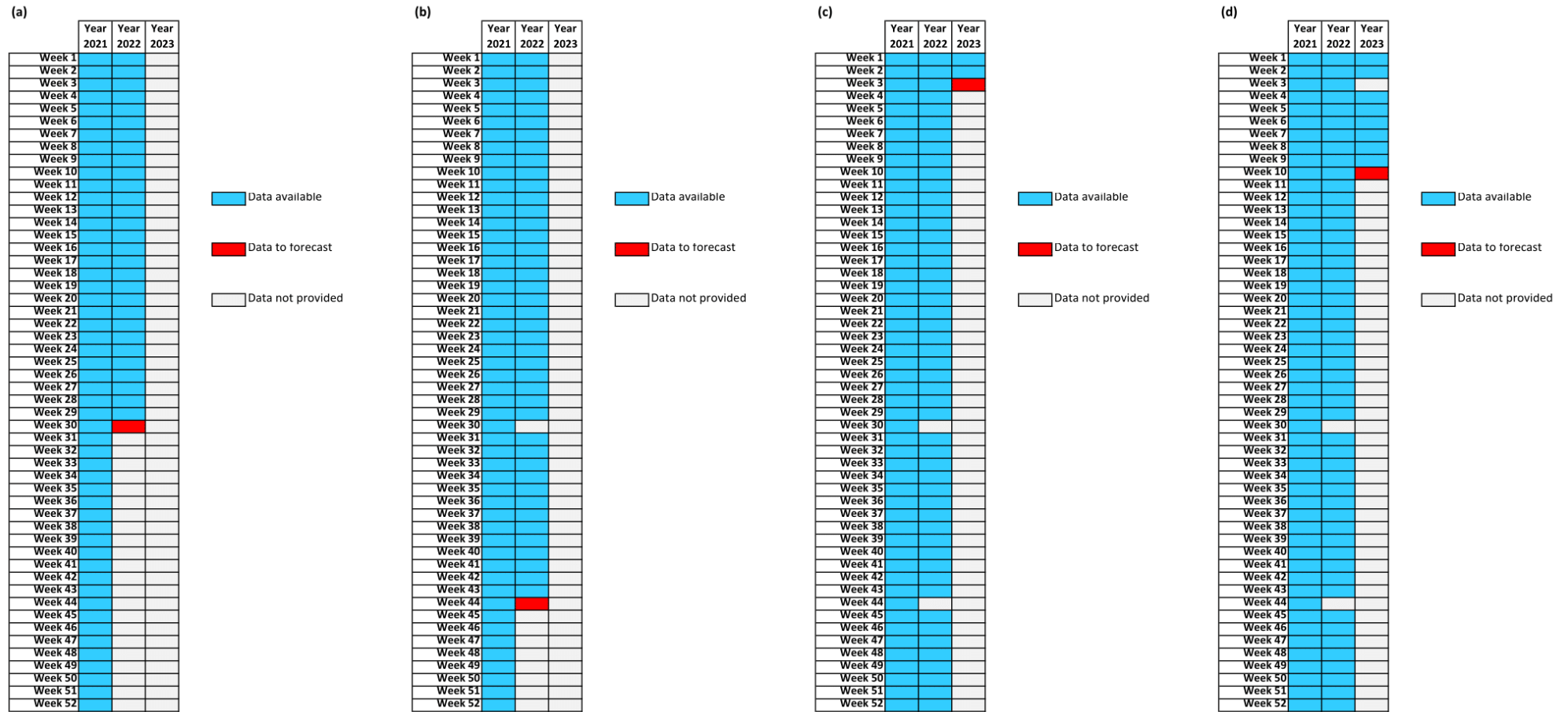


Figure S1. Historical data publication and data-forecast submission deadlines.

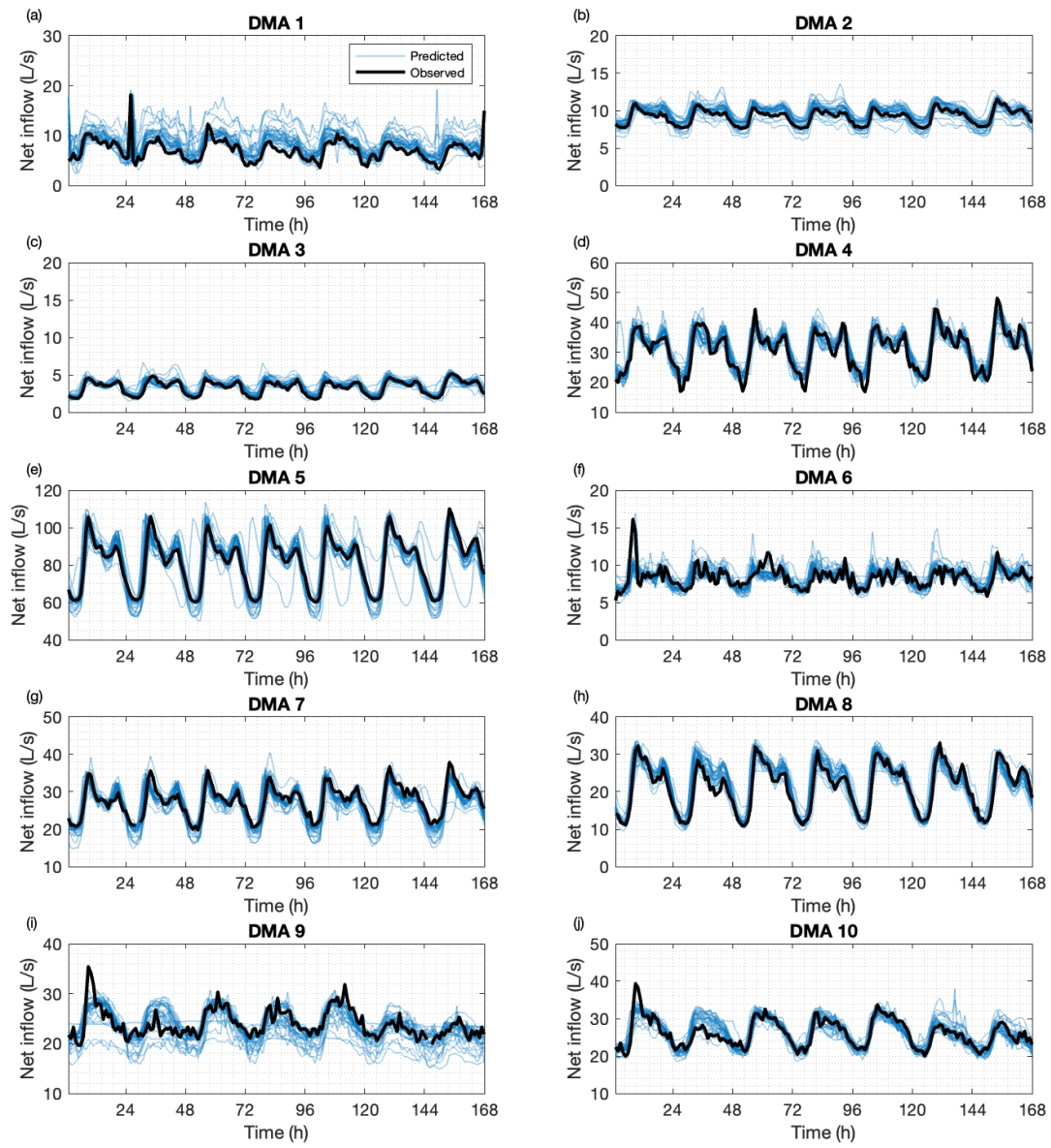


Figure S2. Observed and forecasted inflow in the 10 DMAs during the second evaluation week (W2).

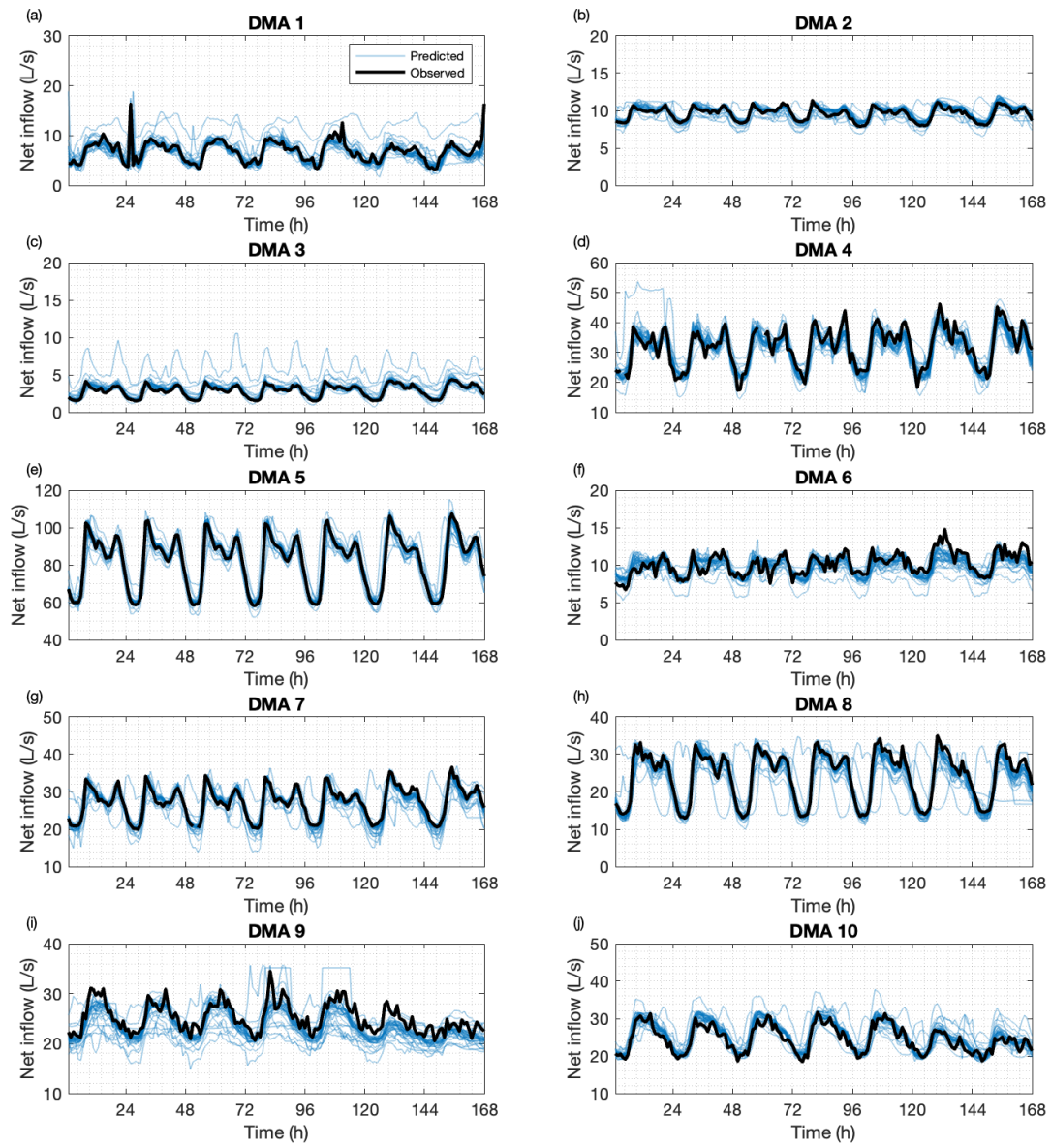


Figure S3. Observed and forecasted inflow in the 10 DMAs during the third evaluation week (W3).

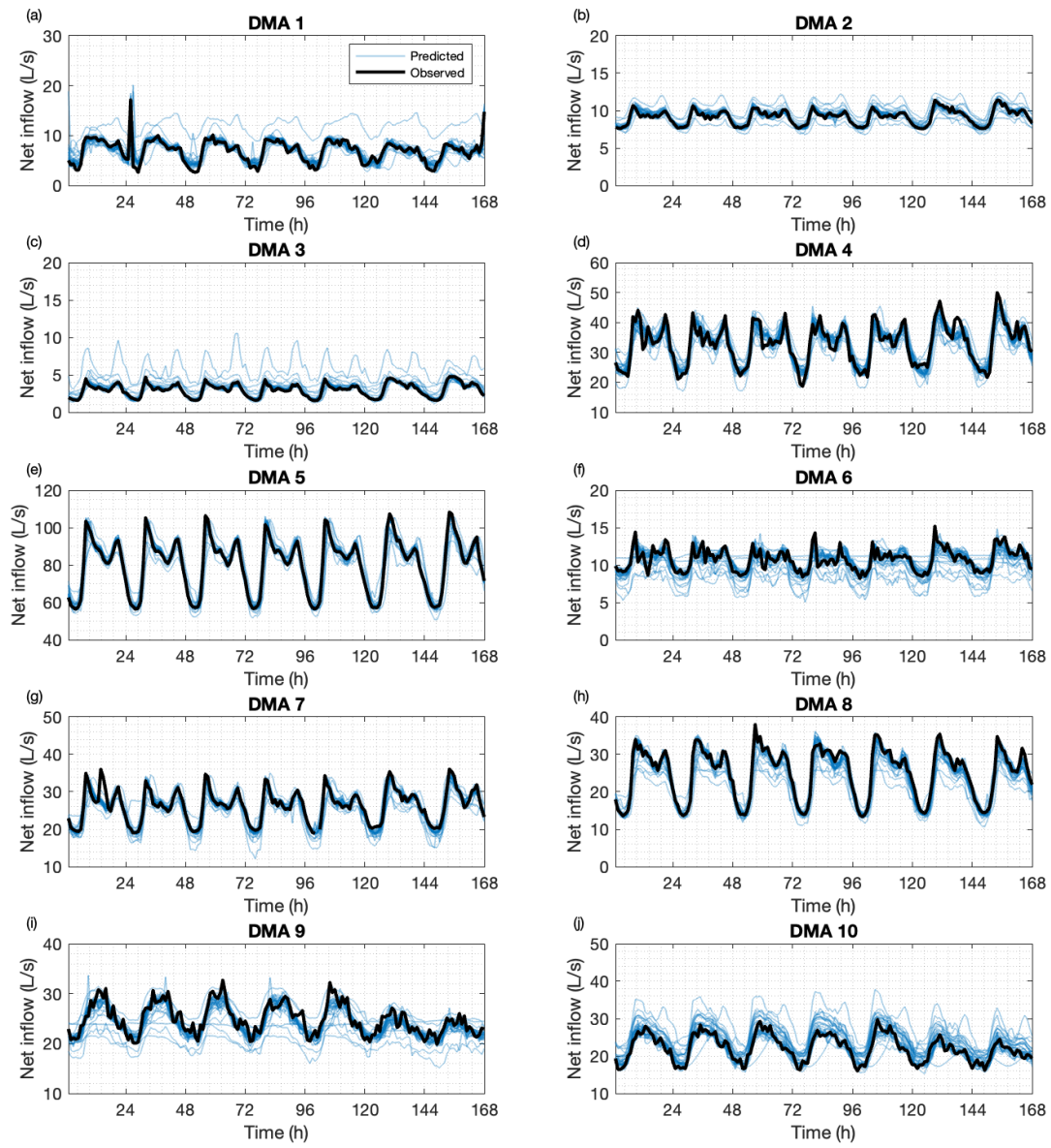


Figure S4. Observed and forecasted inflow in the 10 DMAs during the fourth evaluation week (W4).

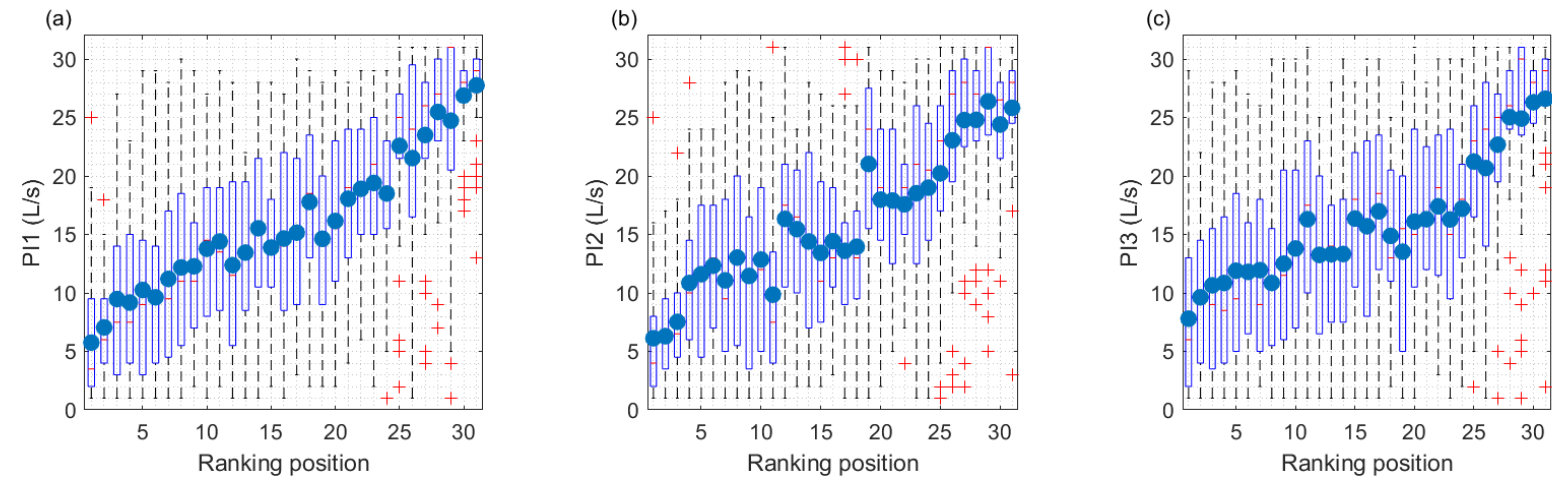


Figure S5. Boxplot of the single rank values and correspondent average rank-value R (blue dot) if PIs are considered individually: (a) $PI1$, (b) $PI2$ and (c) $PI3$.

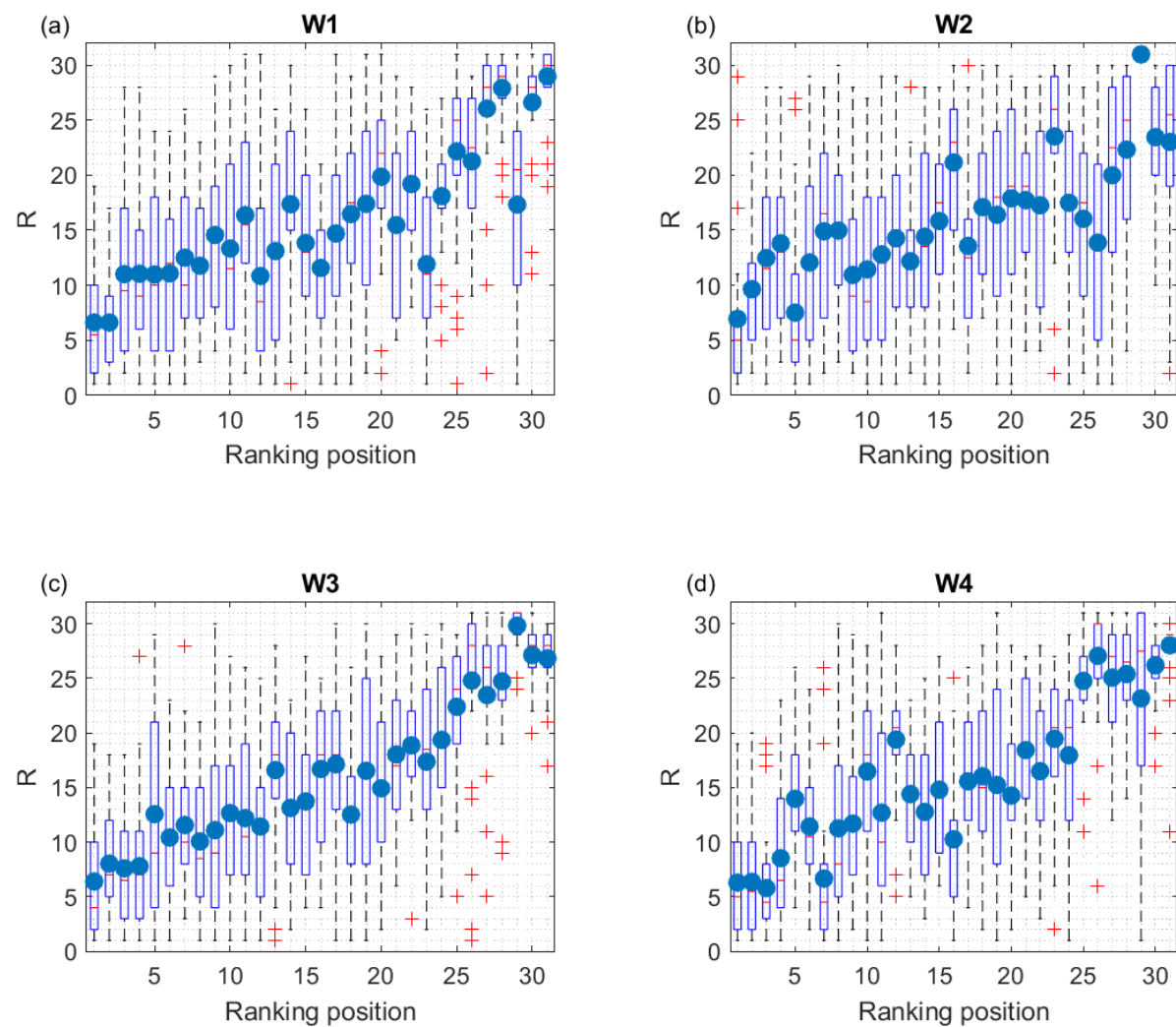


Figure S6. Boxplot of the single rank values if evaluation weeks are considered individually: (a) W1, (b) W2, (c) W3 and (d) W4.

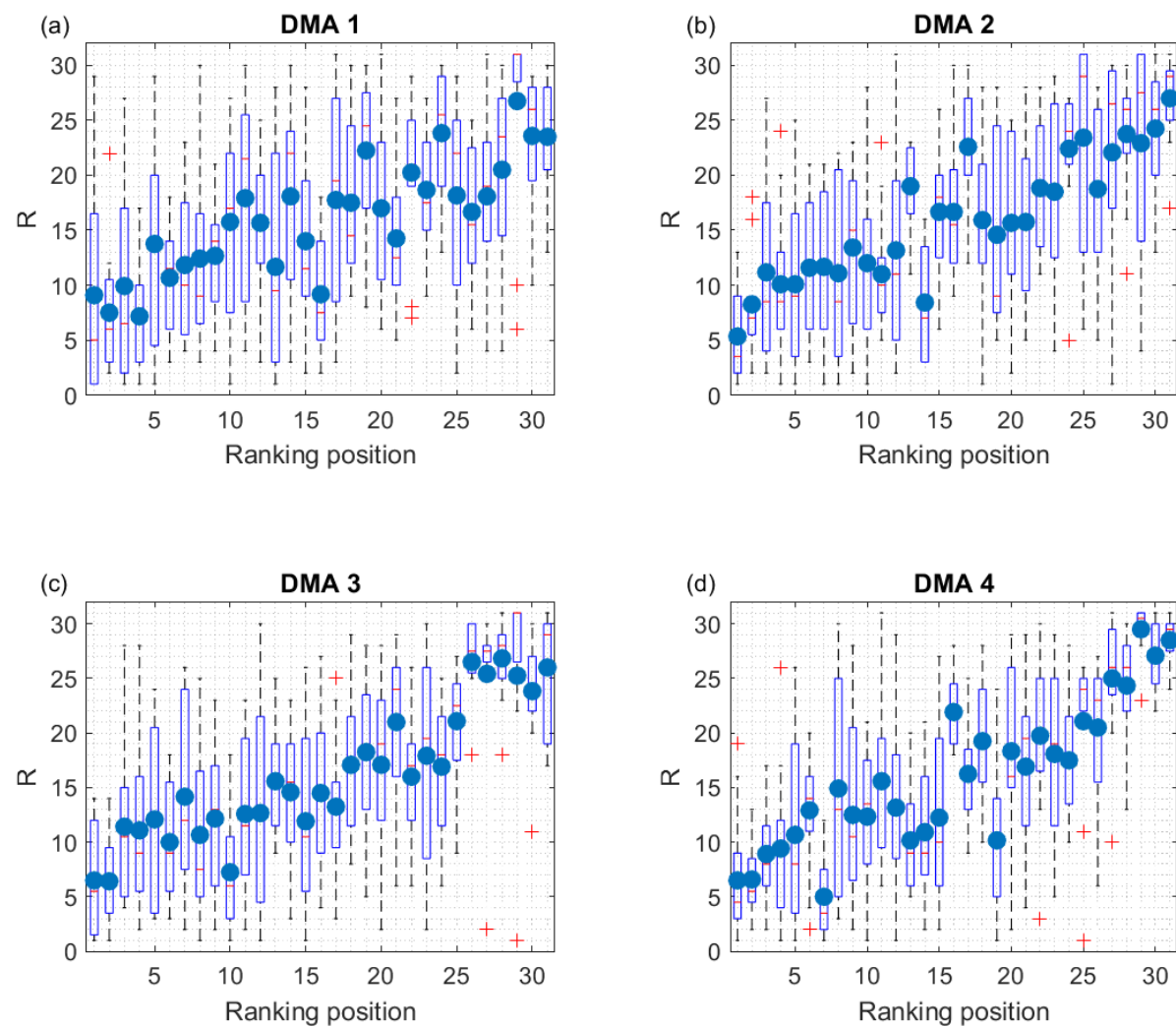


Figure S7. Boxplot of the single rank values if DMAs are considered individually (part I): (a) DMA 1, (b) DMA 2, (c) DMA 3 and (d) DMA 4.

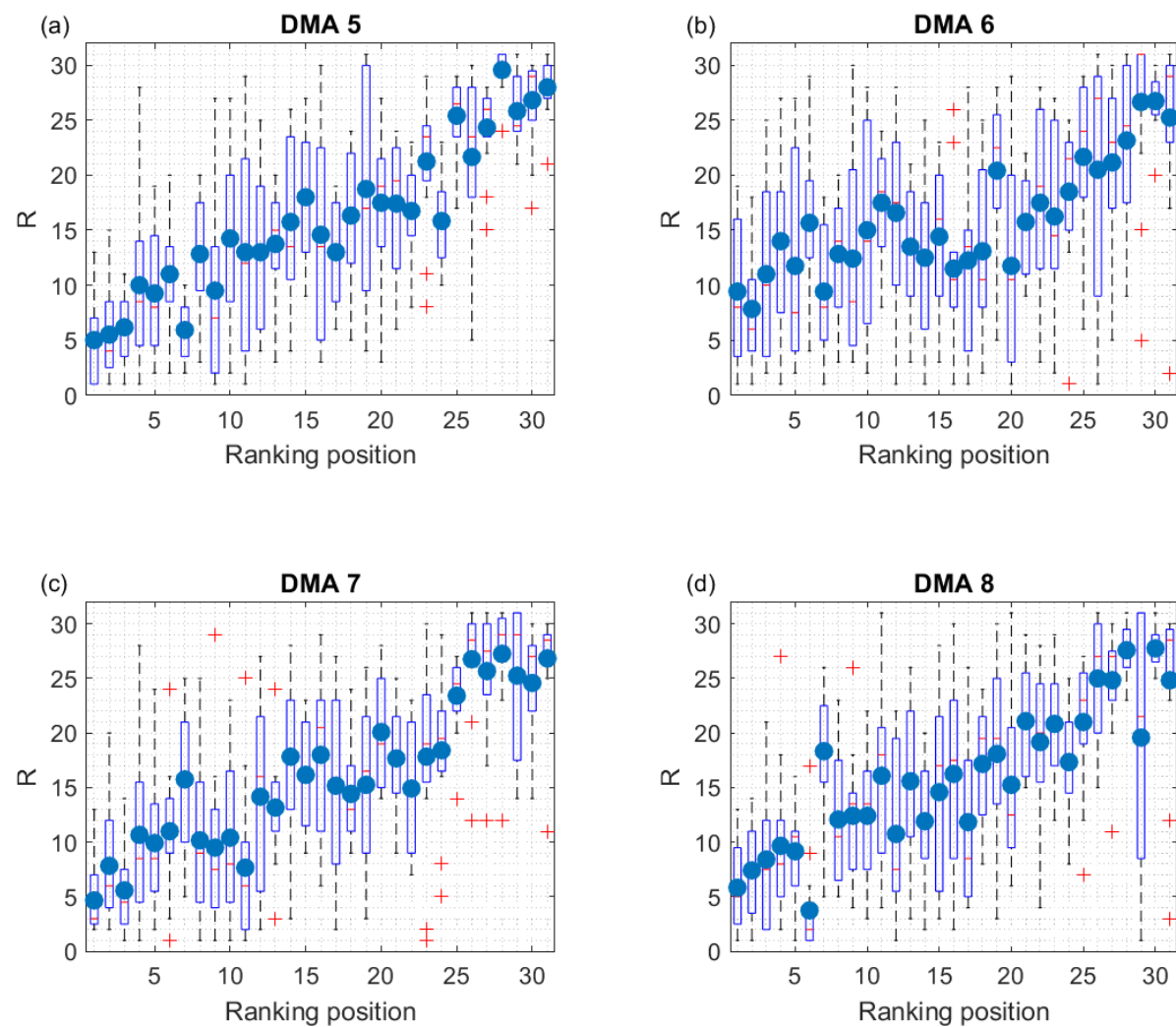


Figure S8. Boxplot of the single rank values if DMAs are considered individually (part II): (a) DMA 5, (b) DMA 6, (c) DMA 7 and (d) DMA 8.

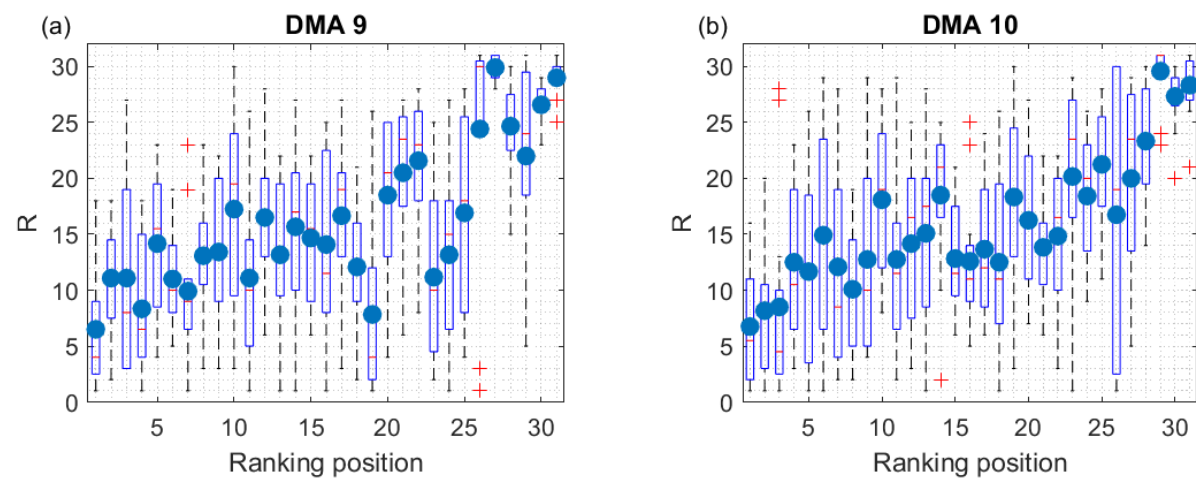


Figure S9. Boxplot of the single rank values if DMAs are considered individually (part III): (a) DMA 9 and (b) DMA 10.

The Battle of Water Demand Forecasting (BWDF)

Datasets – List of contents

- **InflowData.xlsx.** Microsoft Excel® spreadsheet including the hourly net inflow observed in DMAs 1–10 from January 1st, 2021, to March 31st, 2023 (L/s).
- **WeatherData.xlsx.** Microsoft Excel® spreadsheet including hourly values of rainfall depth, air temperature, air humidity, and wind speed observed in the water distribution network including DMAs 1–10 from January 1st, 2021, to March 31st, 2023.
- **CalendarData.pdf.** Working days, holidays, and days with time changes from Central European Time (CET) to Central European Summer Time (CEST) and vice-versa.
- **TeamSolutions.xlsx.** Microsoft Excel® spreadsheet including team forecasts, i.e. predicted net inflow in relation to each DMA and evaluation week. Solutions are organized in 31 datasheets – a datasheet per team – which are sorted based on ranking position (see Figure 4).

January 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
					1	2	3
1	4	5	6	7	8	9	10
2	11	12	13	14	15	16	17
3	18	19	20	21	22	23	24
4	25	26	27	28	29	30	31

May 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
						1	2
18	3	4	5	6	7	8	9
19	10	11	12	13	14	15	16
20	17	18	19	20	21	22	23
21	24	25	26	27	28	29	30
22	31						

September 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
			1	2	3	4	5
36	6	7	8	9	10	11	12
37	13	14	15	16	17	18	19
38	20	21	22	23	24	25	26
39	27	28	29	30			

February 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
5	1	2	3	4	5	6	7
6	8	9	10	11	12	13	14
7	15	16	17	18	19	20	21
8	22	23	24	25	26	27	28

June 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
		1	2	3	4	5	6
23	7	8	9	10	11	12	13
24	14	15	16	17	18	19	20
25	21	22	23	24	25	26	27
26	28	29	30				

October 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
					1	2	3
40	4	5	6	7	8	9	10
41	11	12	13	14	15	16	17
42	18	19	20	21	22	23	24
43	25	26	27	28	29	30	31

March 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
9	1	2	3	4	5	6	7
10	8	9	10	11	12	13	14
11	15	16	17	18	19	20	21
12	22	23	24	25	26	27	28
13	29	30	31				

July 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
				1	2	3	4
27	5	6	7	8	9	10	11
28	12	13	14	15	16	17	18
29	19	20	21	22	23	24	25
30	26	27	28	29	30	31	

November 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
44	1	2	3	4	5	6	7
45	8	9	10	11	12	13	14
46	15	16	17	18	19	20	21
47	22	23	24	25	26	27	28
48	29	30					

April 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
				1	2	3	4
14	5	6	7	8	9	10	11
15	12	13	14	15	16	17	18
16	19	20	21	22	23	24	25
17	26	27	28	29	30		

August 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
							1
31	2	3	4	5	6	7	8
32	9	10	11	12	13	14	15
33	16	17	18	19	20	21	22
34	23	24	25	26	27	28	29
35	30	31					

December 2021							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
			1	2	3	4	5
49	6	7	8	9	10	11	12
50	13	14	15	16	17	18	19
51	20	21	22	23	24	25	26
52	27	28	29	30	31		

	Holiday
	Standard/Summer time

January 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
						1	2
1	3	4	5	6	7	8	9
2	10	11	12	13	14	15	16
3	17	18	19	20	21	22	23
4	24	25	26	27	28	29	30
5	31						

May 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
							1
18	2	3	4	5	6	7	8
19	9	10	11	12	13	14	15
20	16	17	18	19	20	21	22
21	23	24	25	26	27	28	29
22	30	31					

September 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
				1	2	3	4
36	5	6	7	8	9	10	11
37	12	13	14	15	16	17	18
38	19	20	21	22	23	24	25
39	26	27	28	29	30		

February 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
		1	2	3	4	5	6
6	7	8	9	10	11	12	13
7	14	15	16	17	18	19	20
8	21	22	23	24	25	26	27
9	28						

June 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
			1	2	3	4	5
23	6	7	8	9	10	11	12
24	13	14	15	16	17	18	19
25	20	21	22	23	24	25	26
26	27	28	29	30			

October 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
						1	2
40	3	4	5	6	7	8	9
41	10	11	12	13	14	15	16
42	17	18	19	20	21	22	23
43	24	25	26	27	28	29	30
44	31						

March 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
		1	2	3	4	5	6
10	7	8	9	10	11	12	13
11	14	15	16	17	18	19	20
12	21	22	23	24	25	26	27
13	28	29	30	31			

July 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
					1	2	3
27	4	5	6	7	8	9	10
28	11	12	13	14	15	16	17
29	18	19	20	21	22	23	24
30	25	26	27	28	29	30	31

November 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
		1	2	3	4	5	6
45	7	8	9	10	11	12	13
46	14	15	16	17	18	19	20
47	21	22	23	24	25	26	27
48	28	29	30				

April 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
					1	2	3
14	4	5	6	7	8	9	10
15	11	12	13	14	15	16	17
16	18	19	20	21	22	23	24
17	25	26	27	28	29	30	

August 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
31	1	2	3	4	5	6	7
32	8	9	10	11	12	13	14
33	15	16	17	18	19	20	21
34	22	23	24	25	26	27	28
35	29	30	31				

December 2022							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
				1	2	3	4
49	5	6	7	8	9	10	11
50	12	13	14	15	16	17	18
51	19	20	21	22	23	24	25
52	26	27	28	29	30	31	

	Holiday
	Standard/Summer time

January 2023							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
							1
1	2	3	4	5	6	7	8
2	9	10	11	12	13	14	15
3	16	17	18	19	20	21	22
4	23	24	25	26	27	28	29
5	30	31					

February 2023							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
			1	2	3	4	5
6	6	7	8	9	10	11	12
7	13	14	15	16	17	18	19
8	20	21	22	23	24	25	26
9	27	28					

March 2023							
Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
			1	2	3	4	5
10	6	7	8	9	10	11	12
11	13	14	15	16	17	18	19
12	20	21	22	23	24	25	26
13	27	28	29	30	31		Summer time

	Holiday
	Standard/Summer time